Bayes Error

Dated:

Feature $X \in \{0,1,2\}$

Two Classes C_1, C_2

Joint Probabilities given in table $(P(X=x, \text{class}))$:-

x	P(X = x, C1)	P(X=x, C2)	P(X = x)
0	0.10	0.05	0.15
1	0.20	0.10	0.30
2	0.10	0.45	0.55

$\sum P(X=x) = 1.0$

For each x, compute posterior probabilities :-

$$P(C1|x) = \frac{P(x, C1)}{P(x)}, \quad P(C2|x) = \frac{P(x, C2)}{P(x)}$$

Compute :-

$x = 0$:- $P(C1|0) = \frac{0.10}{0.15} = 0.6667$

$P(C2|0) = \frac{0.05}{0.15} = 0.3333$

Bayes picks C1 ; error at $x=0$ min (posteriors) = 0.3333

$x = 1$:- $P(C1|1) = \frac{0.20}{0.30} = 0.6667$

$P(C2|1) = 0.3333$

at $x = 1$ Error = 0.3333

$x = 2$ $P(C1|2) = \frac{0.10}{0.55} \approx 0.1818$

$P(C2|2) = \frac{0.45}{0.55} \approx 0.8182$

Bayes picks C2 , $1 - \max(0.18, 0.81) = 0.1818$

Bayes error overall = $\sum_x P(x) \cdot$ error at x.

↳ = $(0.15)(0.3333) + (0.55)(0.1818) + (0.30)(0.3333) = 0.25$

Bayes error = 0.25 or 25%.

CamScanner

Bayes Error – Additional Practical Discrete Examples

Prepared for study

December 10, 2025

Quick reminder

For discrete features the Bayes classifier at a feature value x chooses the class with the largest posterior

$$P(C_i \mid x) = \frac{P(x, C_i)}{P(x)}.$$

The Bayes (minimum) error is

$$R^* = \sum_x P(x) \min_i\{P(C_i \mid x)\},$$

i.e. sum over each x of the probability of x times the smaller posterior at x.

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1 Example 4: Email spam detection (word-count feature)

Problem setup:

- Feature $X \in \{0, 1, 2, 3\}$ = number of "suspicious" words in an email (0 = none, 3 = many).
- Two classes: C_{spam} and C_{ham} (non-spam).
- Joint probabilities $P(X = x, C)$ (empirical):

x	$P(x, \text{spam})$	$P(x, \text{ham})$	$P(x) = P(x, \text{spam}) + P(x, \text{ham})$
0	0.12	0.28	0.40
1	0.10	0.20	0.30
2	0.08	0.08	0.16
3	0.10	0.04	0.14

Table 1: Joint and marginal probabilities (Email example).

(Verify: $0.40 + 0.30 + 0.16 + 0.14 = 1.00$.)

Step 1 – Posterior probabilities:

$$P(\text{spam} \mid x) = \frac{P(x, \text{spam})}{P(x)}, \qquad P(\text{ham} \mid x) = \frac{P(x, \text{ham})}{P(x)}.$$

Compute precisely:

$$\begin{aligned}
x = 0: &\quad P(\text{spam} \mid 0) = \frac{0.12}{0.40} = 0.3000, \quad P(\text{ham} \mid 0) = \frac{0.28}{0.40} = 0.7000. \\
x = 1: &\quad P(\text{spam} \mid 1) = \frac{0.10}{0.30} = 0.333333\ldots \approx 0.3333, \quad P(\text{ham} \mid 1) = 0.666666\ldots \approx 0.6667. \\
x = 2: &\quad P(\text{spam} \mid 2) = \frac{0.08}{0.16} = 0.5000, \quad P(\text{ham} \mid 2) = 0.5000. \\
x = 3: &\quad P(\text{spam} \mid 3) = \frac{0.10}{0.14} \approx 0.7142857 \approx 0.7143, \quad P(\text{ham} \mid 3) \approx 0.2857143 \approx 0.2857.
\end{aligned}$$

Bayes decision rule (per x):

- $x = 0$: choose **ham** (0.7000 vs 0.3000).
- $x = 1$: choose **ham** (0.6667 vs 0.3333).
- $x = 2$: tie (0.5000 vs 0.5000). Break ties by a rule — here choose **ham**.
- $x = 3$: choose **spam** (0.7143 vs 0.2857).

Step 2 – Bayes error (overall): error contribution at each x is $P(x) \cdot \min\{P(\text{spam} \mid x), P(\text{ham} \mid x)\}$.

$$\begin{aligned}
\text{Error} &= 0.40 \cdot 0.3000 + 0.30 \cdot 0.3333 + 0.16 \cdot 0.5000 + 0.14 \cdot 0.2857 \\
&= 0.1200 + 0.1000 + 0.0800 + 0.0400 \\
&= 0.3400.
\end{aligned}$$

x	$P(x, \text{spam})$	$P(x, \text{ham})$	$P(x) = P(x, \text{spam}) + P(x, \text{ham})$
0	0.12	0.28	0.40
1	0.10	0.20	0.30
2	0.08	0.08	0.16
3	0.10	0.04	0.14

Table 1: Joint and marginal probabilities (Email example).

(Verify: $0.40 + 0.30 + 0.16 + 0.14 = 1.00$.)

Step 1 – Posterior probabilities:

$$P(\text{spam} \mid x) = \frac{P(x, \text{spam})}{P(x)}, \qquad P(\text{ham} \mid x) = \frac{P(x, \text{ham})}{P(x)}.$$

Compute precisely:

$$x = 0: \quad P(\text{spam} \mid 0) = \frac{0.12}{0.40} = 0.3000, \quad P(\text{ham} \mid 0) = \frac{0.28}{0.40} = 0.7000.$$
$$x = 1: \quad P(\text{spam} \mid 1) = \frac{0.10}{0.30} = 0.333333\ldots \approx 0.3333, \quad P(\text{ham} \mid 1) = 0.666666\ldots \approx 0.6667.$$
$$x = 2: \quad P(\text{spam} \mid 2) = \frac{0.08}{0.16} = 0.5000, \quad P(\text{ham} \mid 2) = 0.5000.$$
$$x = 3: \quad P(\text{spam} \mid 3) = \frac{0.10}{0.14} \approx 0.7142857 \approx 0.7143, \quad P(\text{ham} \mid 3) \approx 0.2857143 \approx 0.2857.$$

Bayes decision rule (per x):

- $x = 0$: choose **ham** (0.7000 vs 0.3000).
- $x = 1$: choose **ham** (0.6667 vs 0.3333).
- $x = 2$: tie (0.5000 vs 0.5000). Break ties by a rule — here choose **ham**.
- $x = 3$: choose **spam** (0.7143 vs 0.2857).

Step 2 – Bayes error (overall): error contribution at each x is $P(x) \cdot \min\{P(\text{spam} \mid x), P(\text{ham} \mid x)\}$.

$$\begin{aligned} \text{Error} &= 0.40 \cdot 0.3000 + 0.30 \cdot 0.3333 + 0.16 \cdot 0.5000 + 0.14 \cdot 0.2857 \\ &= 0.1200 + 0.1000 + 0.0800 + 0.0400 \\ &= 0.3400. \end{aligned}$$

Result: Bayes error $R^* \approx 0.34$ (34%).

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2 Example 5: Credit approval (risk score)

Problem setup:

- Feature $X \in \{\text{Low}, \text{Medium}, \text{High}\}$ is a discrete credit risk score.
- Two classes: $C_A = \text{Approve}$ and $C_R = \text{Reject}$.
- Joint probabilities (small dataset):

X	$P(X, C_A)$	$P(X, C_R)$	$P(X) = P(X, C_A) + P(X, C_R)$
Low	0.35	0.05	0.40
Medium	0.20	0.15	0.35
High	0.03	0.22	0.25

Table 2: Joint and marginal probabilities (Credit example).

(Verify: $0.40 + 0.35 + 0.25 = 1.00$.)
Step 1 – Posteriors:

$$
\begin{aligned}
&X = \text{Low}: \quad P(A \mid \text{Low}) = \frac{0.35}{0.40} = 0.8750, \quad P(R \mid \text{Low}) = 0.1250. \\
&X = \text{Medium}: \quad P(A \mid \text{Medium}) = \frac{0.20}{0.35} = 0.5714286 \approx 0.5714, \\
&\qquad\qquad P(R \mid \text{Medium}) = 0.4285714 \approx 0.4286. \\
&X = \text{High}: \quad P(A \mid \text{High}) = \frac{0.03}{0.25} = 0.1200, \quad P(R \mid \text{High}) = 0.8800.
\end{aligned}
$$

Bayes decision rule:

- Low: choose **Approve** (0.8750 vs 0.1250).
- Medium: choose **Approve** (0.5714 vs 0.4286).
- High: choose **Reject** (0.8800 vs 0.1200).

Step 2 – Bayes error:

$$\begin{aligned}
\text{Error} &= 0.40 \cdot 0.1250 + 0.35 \cdot 0.4286 + 0.25 \cdot 0.1200 \\
&= 0.0500 + 0.1500 + 0.0300 \\
&= 0.2300.
\end{aligned}$$

Result: Bayes error $R^* \approx 0.23$ (23%).

Practical notes:

- These worked examples use small empirical joint-probability tables (often estimated from counts). In practice you would estimate $P(x, C_i)$ from training data counts (possibly with smoothing).
- If classes have unequal misclassification costs, the Bayes decision compares expected costs rather than simply choosing the higher posterior.
- Ties in posterior probabilities are rare in real-valued features; if they do occur, choose a tie-breaking rule consistent with domain risk (e.g., prefer safer class).